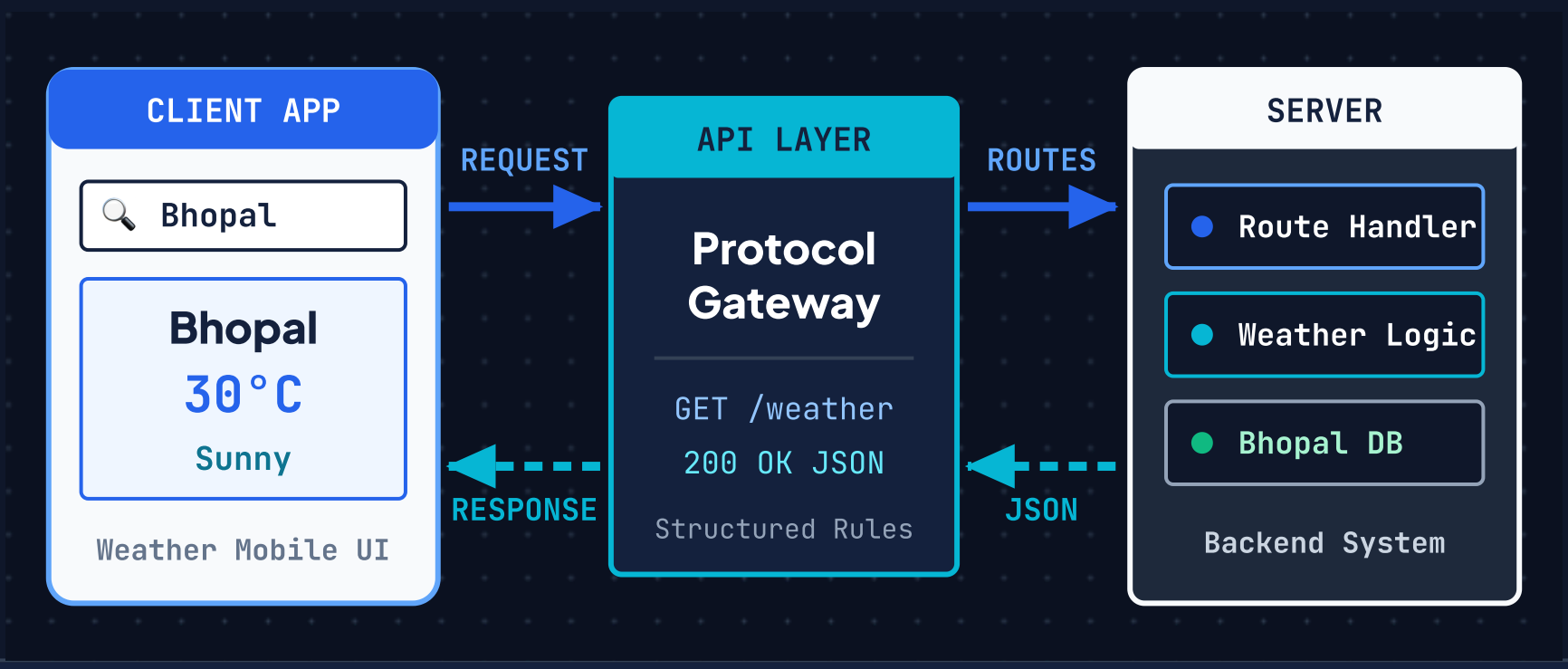
BACKEND COMMUNICATION MECHANICS

HOW DOES AN API ACTUALLY WORK?

> "Understanding the journey from request to response."

FIG 03.1 // THE REQUEST-RESPONSE ROUND TRIP — CLIENT ⇄ API ⇄ SERVER



→ OUTGOING REQUEST (Client → Server)

◄ - - RETURNING RESPONSE (Server → Client)

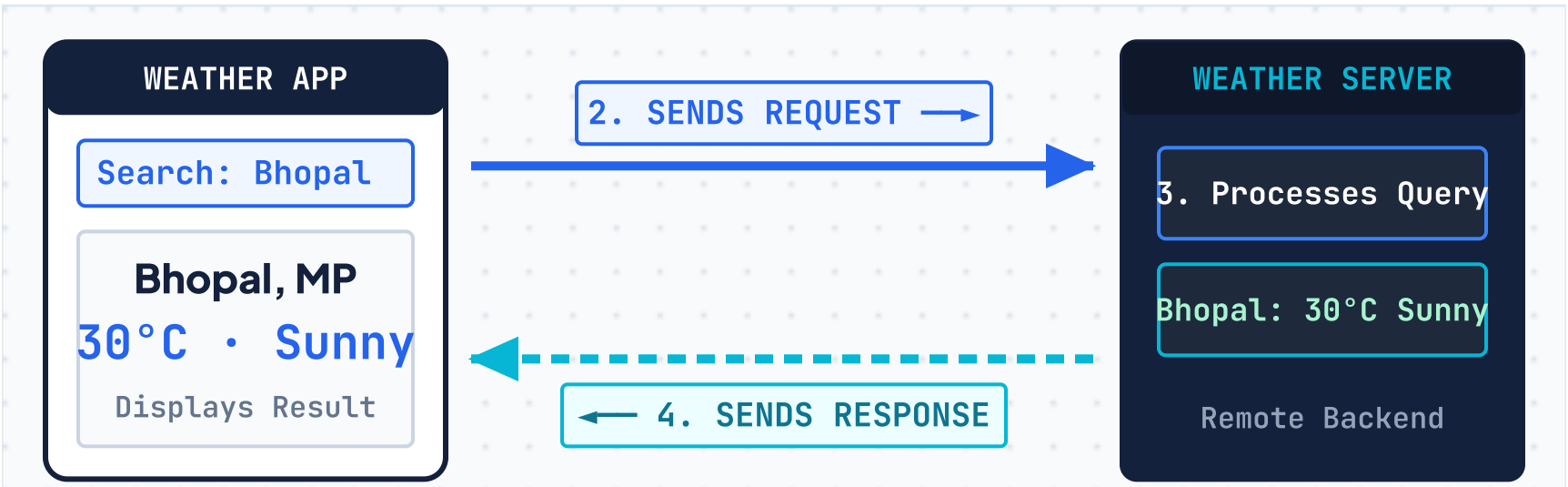
01 // END-TO-END OVERVIEW

Every API interaction starts with a request.

You open a weather application and search for **Bhopal**. To show you the forecast, the application communicates with a remote weather server.

FIG 03.2 // BHOPAL WEATHER LOOKUP ROUND-TRIP

CLIENT ⇄ WEATHER SERVER



01	You search for **Bhopal** in the weather application.
02	The application sends a **request** to the server.
03	The weather server **processes** the incoming request.
04	The server sends a **response** containing weather data.
05	The application **displays** the weather on your screen.

Request goes to the server. Response comes back to the client.

02 // OUTGOING MESSAGE

First, the client asks for something.

A **request** is a message sent by a client to a server asking it to perform an action or provide information.

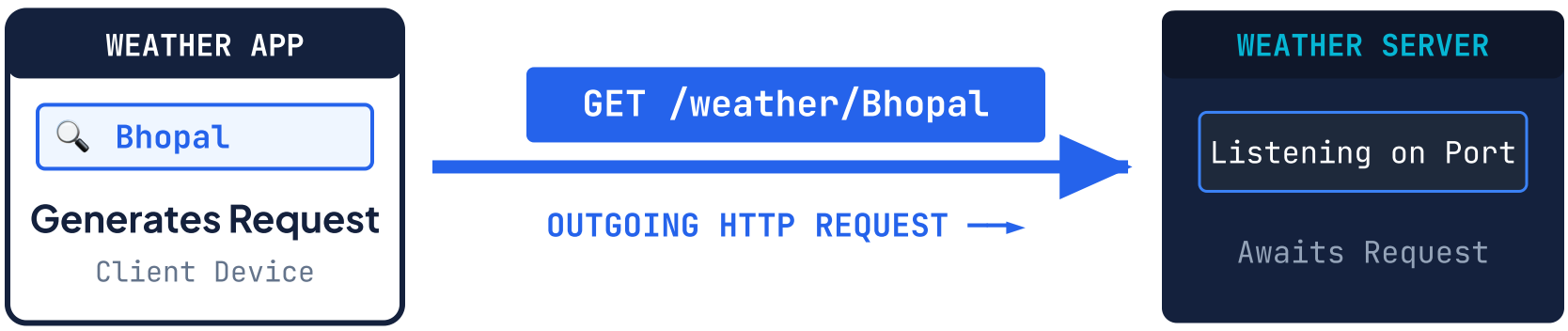
PLAIN-ENGLISH INTENT (WEATHER EXAMPLE)

“I want the weather information for Bhopal.”

SIMPLIFIED HTTP REQUEST LINE — WEB API PROTOCOL: HTTP



FIG 03.3 // CLIENT DISPATCHING THE HTTP REQUEST — DIRECTION: CLIENT → SERVER



03 // TECHNICAL INSPECTION

A request contains more than just a question.

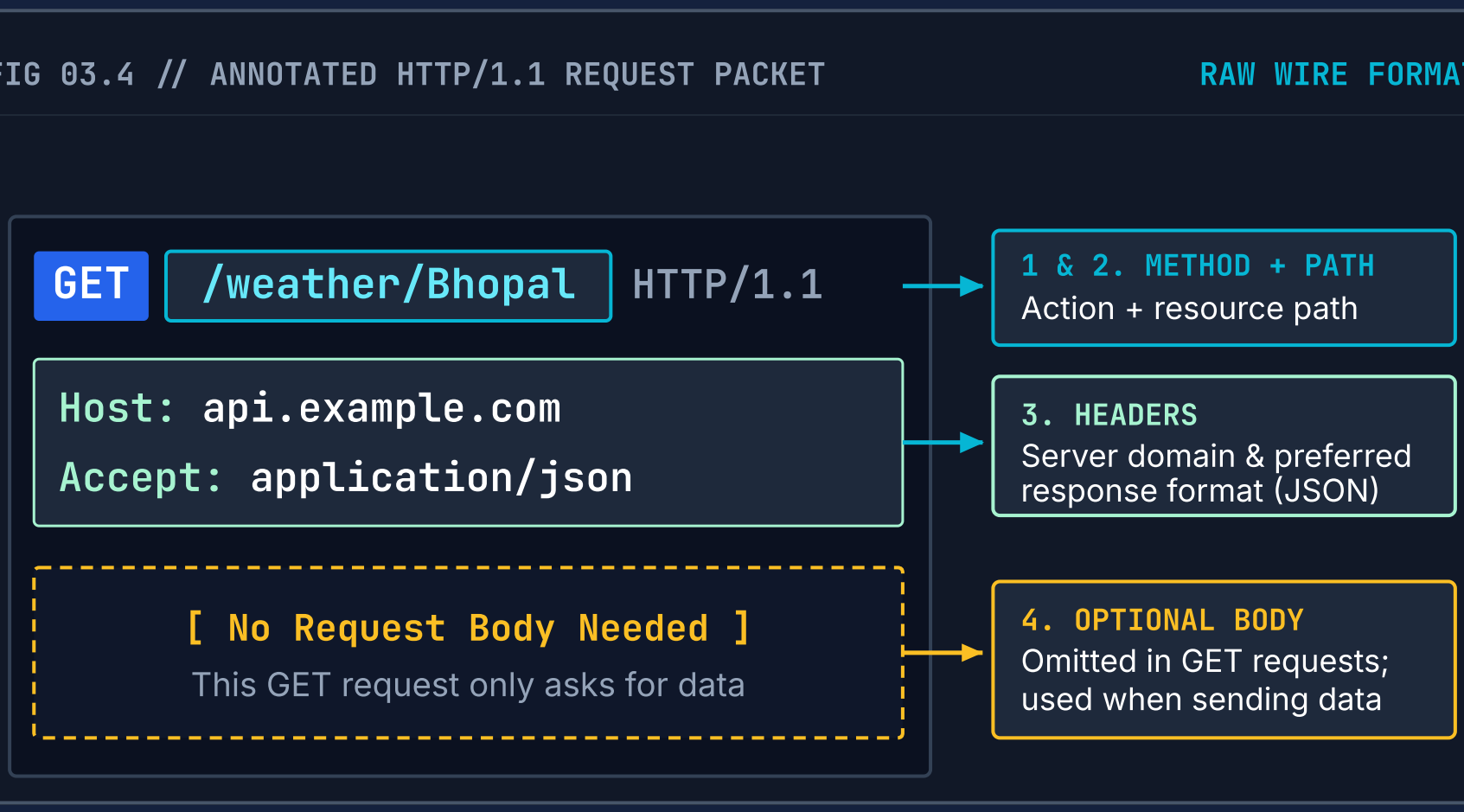
01 // HTTP METHOD
What action is being requested?

02 // URL (HOST + PATH)
Where is the request going?

03 // HEADERS
Additional request information.

04 // BODY (OPTIONAL)
Optional data sent with request.

FIG 03.4 // ANNOTATED HTTP/1.1 REQUEST PACKET — RAW WIRE FORMAT



IMPORTANT TECHNICAL NOTE

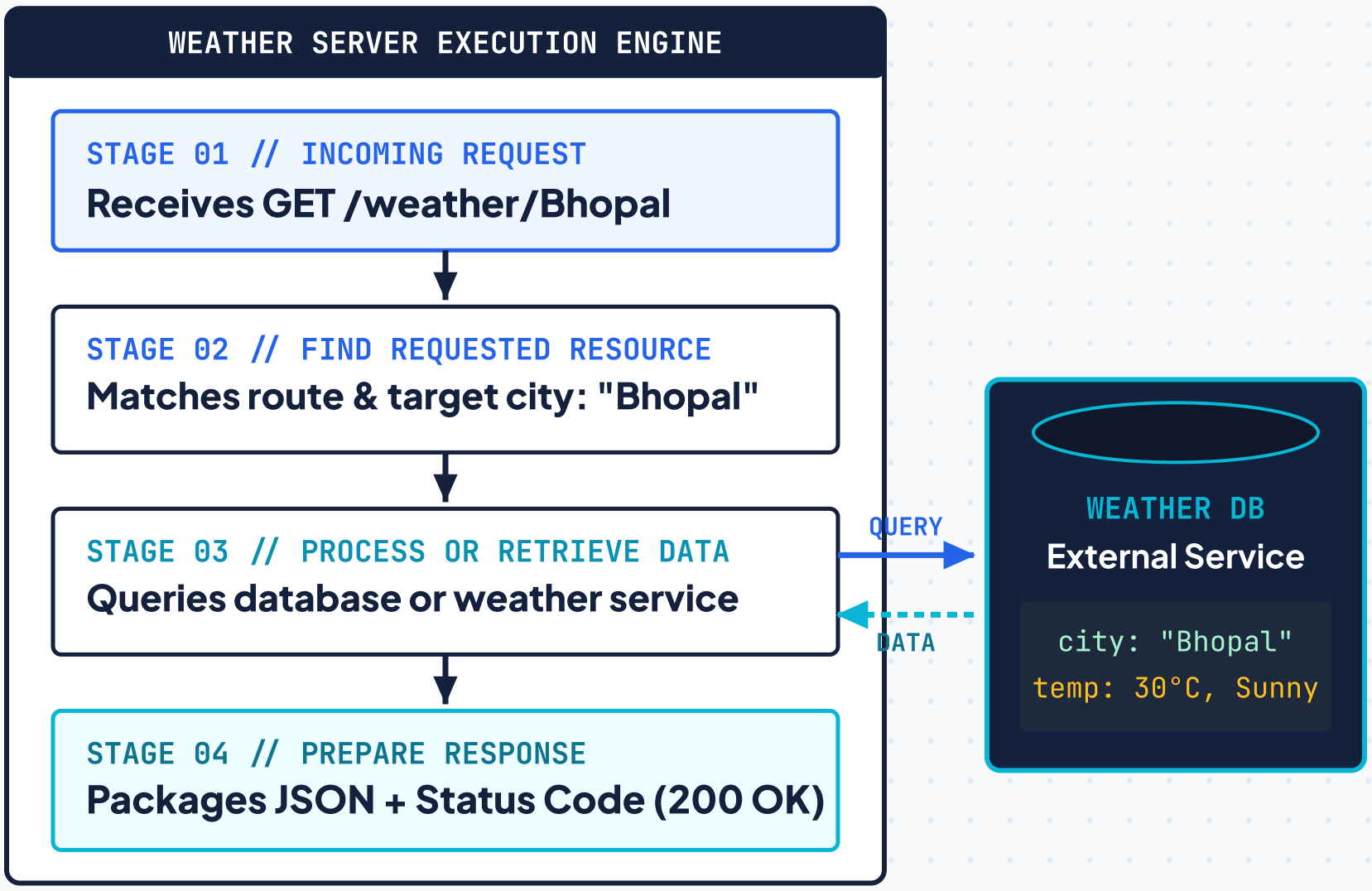
Not every request uses all four parts: a **request body is optional** (this `GET` example does not need one), and **headers** depend on the situation.

04 // INSIDE THE BACKEND

What happens after the server receives it?

The server receives the incoming request and follows a structured sequence to handle it:

FIG 03.5 // SERVER-SIDE PROCESSING PIPELINE WEATHER SERVER + DATABASE LOOKUP



To obtain the required information, the server often queries a **database**, runs Python logic, or communicates with another external service.

05 // RETURNING THE RESULT

The server sends the result back.

A **response** is the message the server sends to the client after processing a request. Web APIs commonly return structured **JSON** data:

HTTP RESPONSE BODY (JSON PAYLOAD) — CONTENT-TYPE: APPLICATION/JSON

```
{
  "city": "Bhopal",
  "temperature": 30,
  "condition": "Sunny"
}
```

FIG 03.6 // SERVER RETURNING JSON DATA TO THE CLIENT — DIRECTION: SERVER → CLIENT



01 // SERVER
Returns structured data in the response.

02 // CLIENT
Receives and reads that structured data.

03 // DISPLAY
Uses the values to display the weather.

06 // RESULT INDICATORS

How does the client know whether the request succeeded?

The server includes a three-digit **HTTP status code** in its response to indicate the exact result of processing the request.

200 [2XX SUCCESS]

OK — Request succeeded

✓ VERIFIED

The server found Bhopal's weather and returned the data normally.

201 [2XX CREATED]

Created — Resource created

+ NEW RECORD

A new item (such as a saved city or user account) was successfully stored.

404 [4XX CLIENT]

Not Found — Resource not found

? MISSING PATH

The requested URL or city name does not exist on the server.

500 [5XX SERVER]

Internal Server Error

! SERVER FAULT

The request was valid, but something failed inside the server's own code or database.

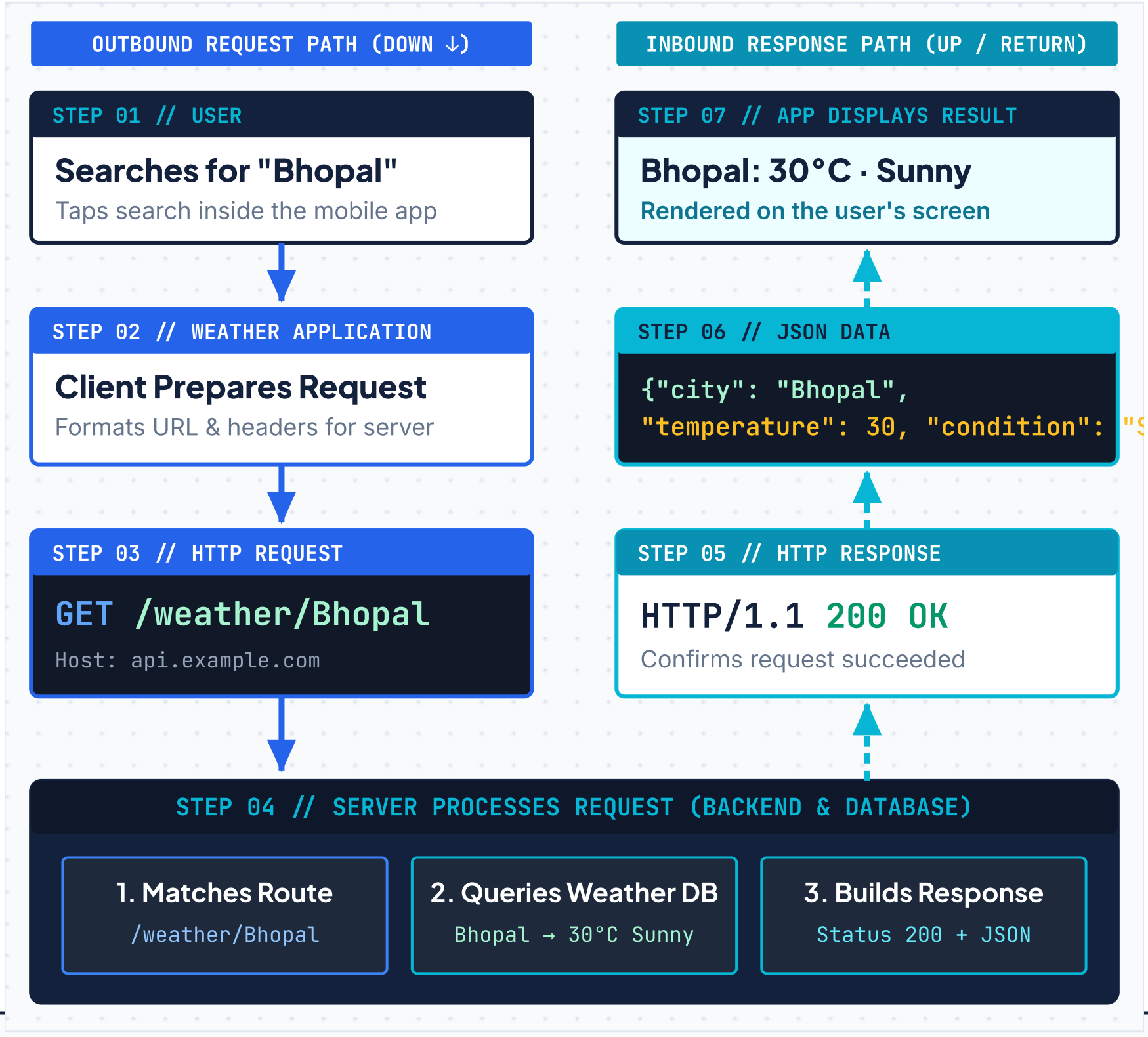
Note: There are many more status codes grouped into `1xx–5xx` families, which we will cover in detail in a dedicated part.

07 // END-TO-END ARCHITECTURAL TRACE

Let's follow one complete API interaction.

FIG 03.7 // COMPLETE BHOPAL WEATHER REQUEST-RESPONSE LIFECYCLE

7-STAGE SEQUENCE



08 // MODULE 03 REFERENCE SUMMARY

You now understand the request–response cycle.

TERM	MEANING
Request	Message sent by the client
HTTP method	Type of action requested
URL	Destination of the request
Headers	Additional request information
Body	Optional data sent with the request
Server	Processes the request
Response	Message returned by the server
Status code	Indicates the result

An API is not magic. It is structured communication between software systems.

COMING NEXT // PART 04

PART 04 — Understanding HTTP Methods: GET, POST, PUT, PATCH and DELETE

Save this carousel for your backend development journey.